University of East Anglia
COMPUTING SCIENCES

Advanced Web Development

Lab 0.2. Using CMP student remote Git repository

Learning Objectives:

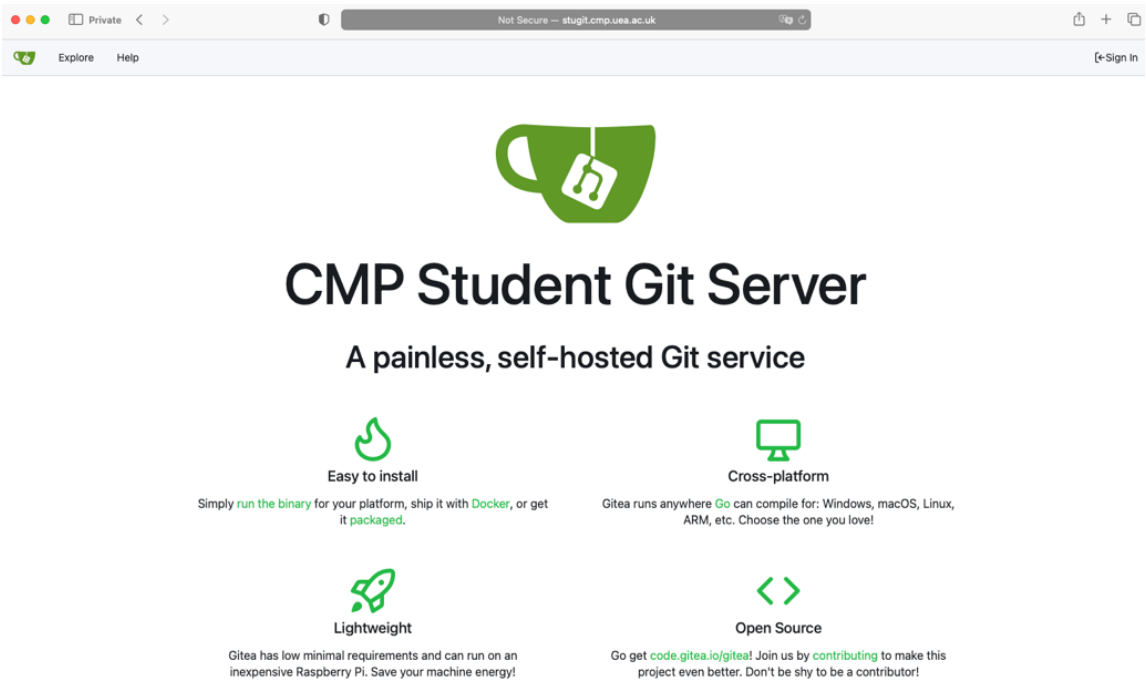
- Practising git commands
- Connecting to remote git server
- Committing files to a remote git server
- Setting up a group repo

Server information:

Hostname: http://stugit.cmp.uea.ac.uk/

Username: to_be_assigned

Password: to_be_assigned



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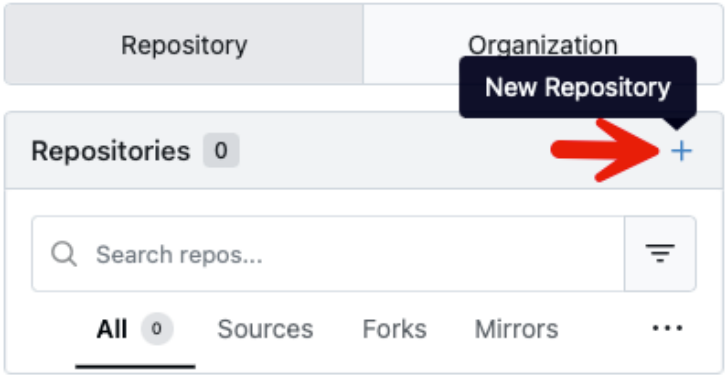
Creating a repository

Follow these steps:

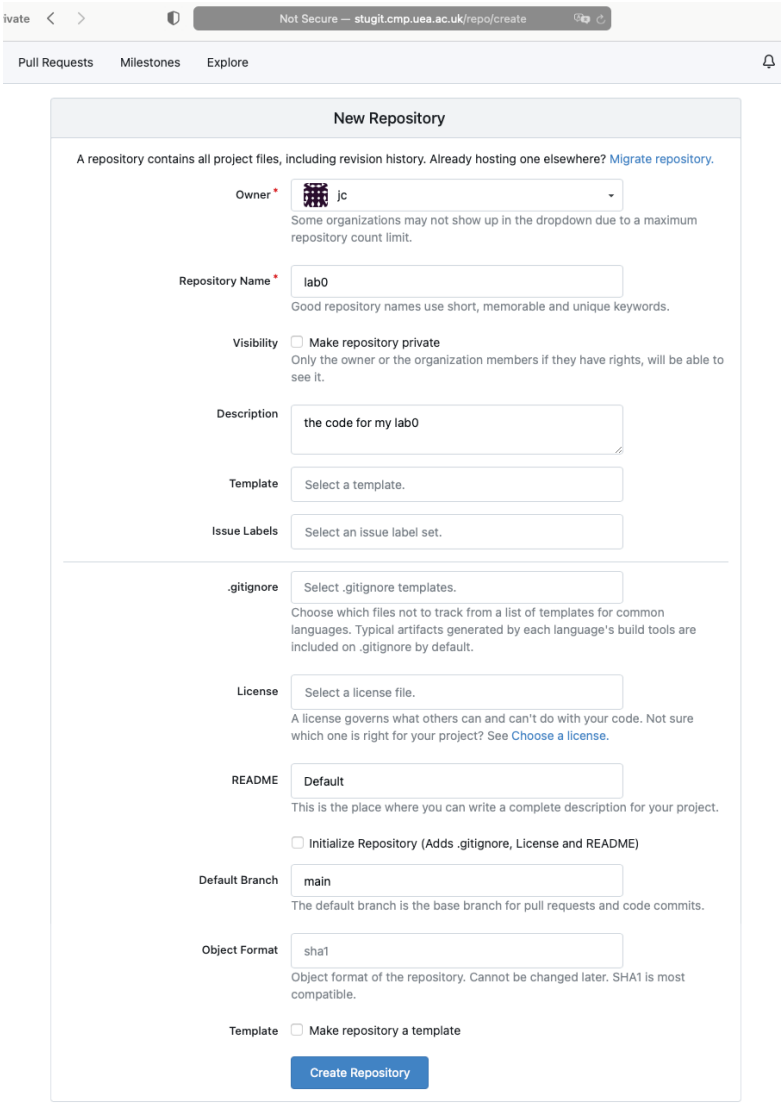
1. On your browser, navigate to http://stugit.cmp.uea.ac.uk/
2. Login using your credentials (note: this will be provided at the start of the term)
3. Once login, you will see a page similar to the following:



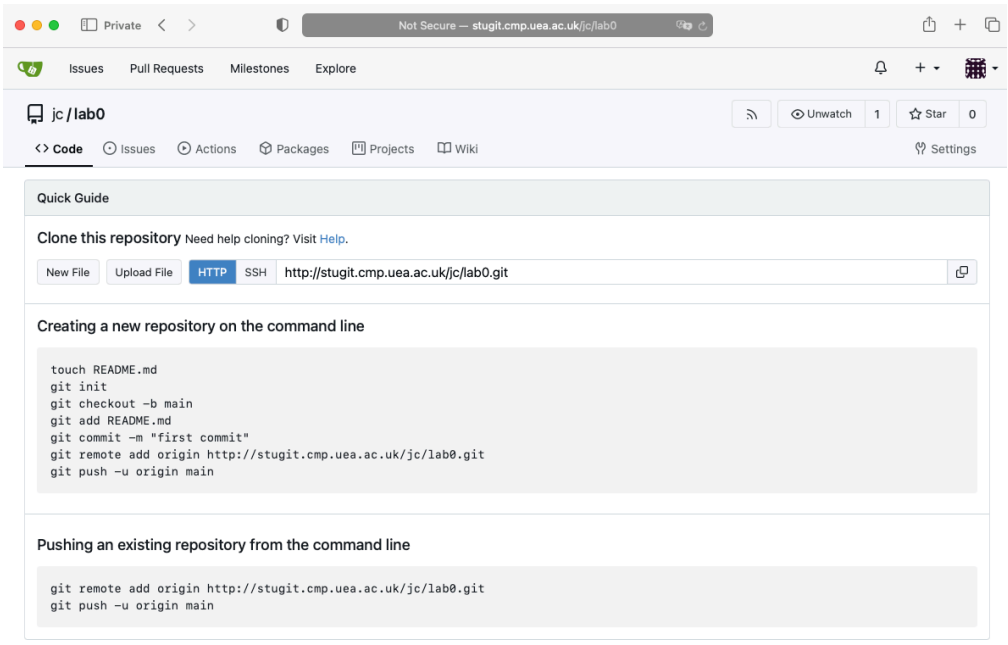
4. Click the '+' to create a new repository



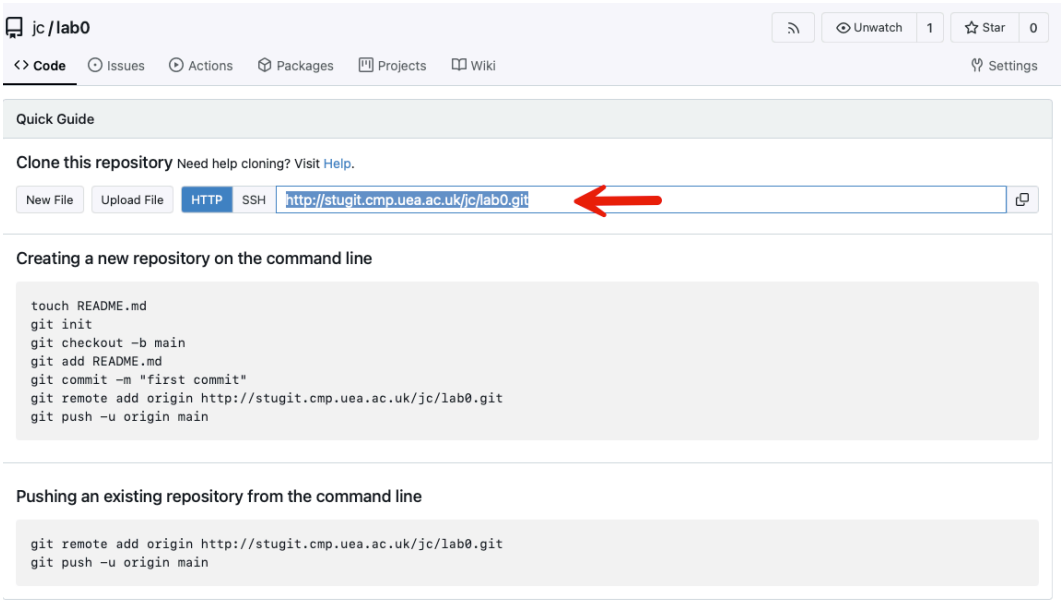
5. On the ‘new repository” page, enter the details for this repo, and click “Create Repository”. Below is an example of the details for my lab0 repo for your information:



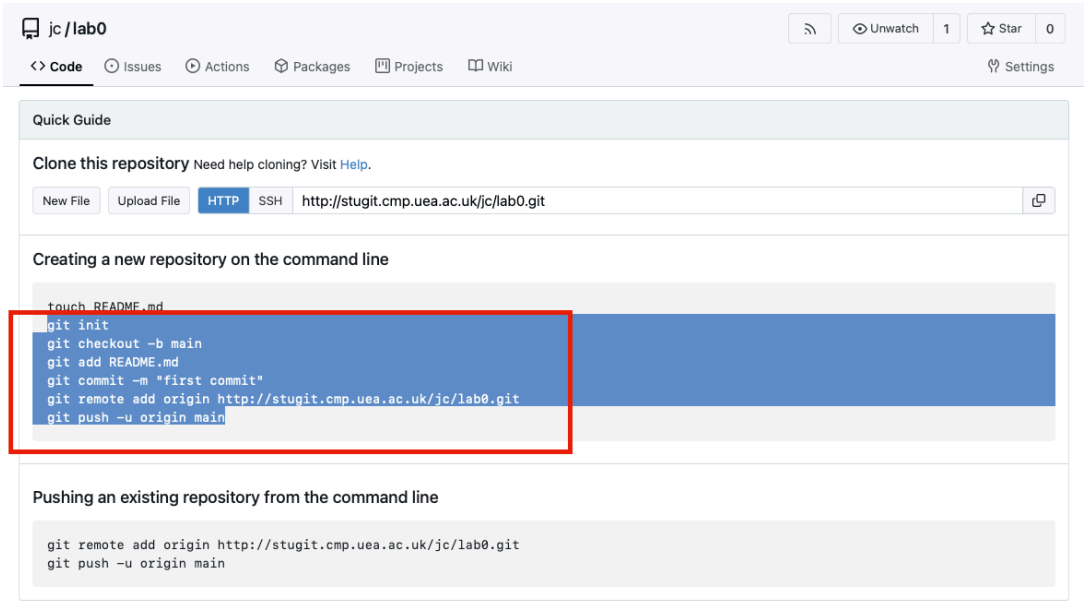
6. You will then see a page similar to the one showed below:



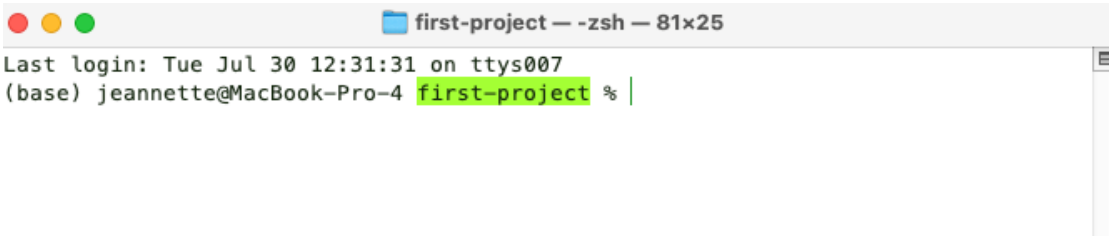
7. You can find the git URL here:



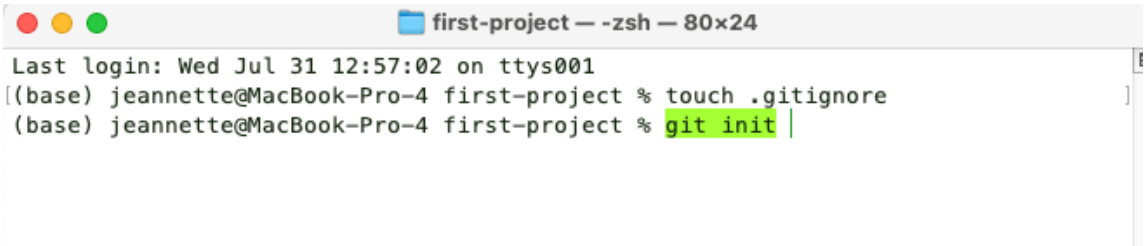
8. Copy these useful commands and store it somewhere, as you will need it later. We can **ignore** the '`touch README.md`" command, as we have got this file when we created our project (created by vite).



9. The next task is to sync this repo to your project folder. Using a terminal window (use CMP PowerShell if you are using a lab machine), navigate to your lab folder. You can see from below, I am using a terminal window (on my mac) and I am in my lab root folder, this folder is called "first-project":

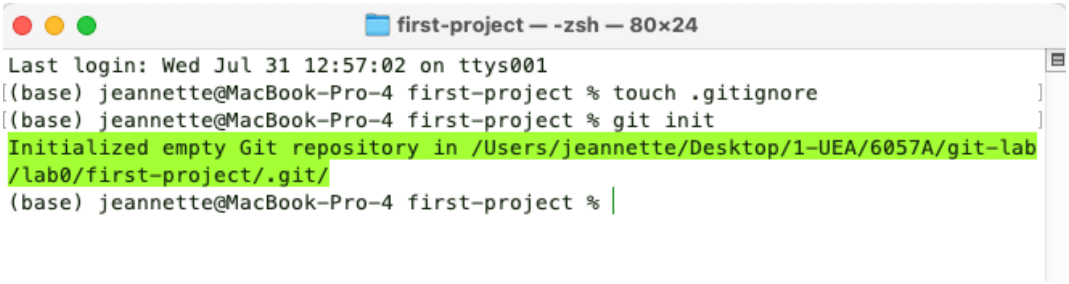


10. On the terminal window, type '`git init`':

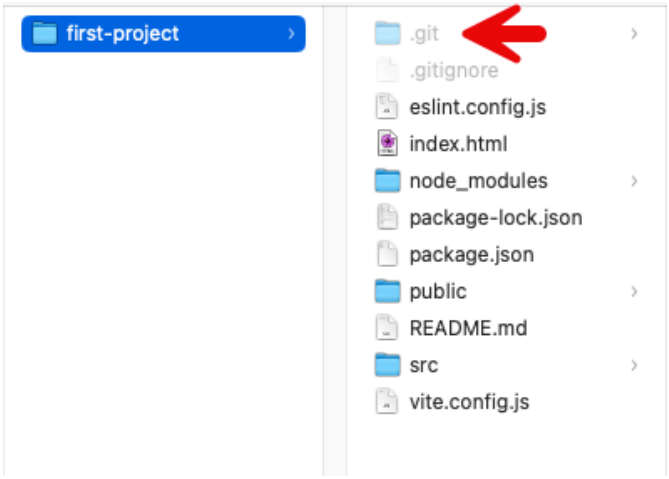


Note: if you have not got `.gitignore` file created in your project root directory, you can use the command '`touch .gitignore`' to create an empty file. You can then specify which file/s and folders to be ignored, such that they stay local, and will not be push to git repo).

11. You will see that a hidden folder called '`.git`' is created inside this folder:



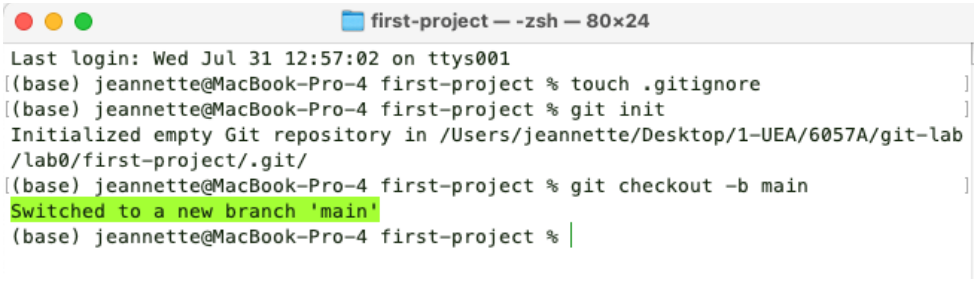
You can double check this hidden folder in the windows view as well:



12. Next, we switch to branch 'main' by typing '`git checkout -b main`'



13. You will see the output similar to below:



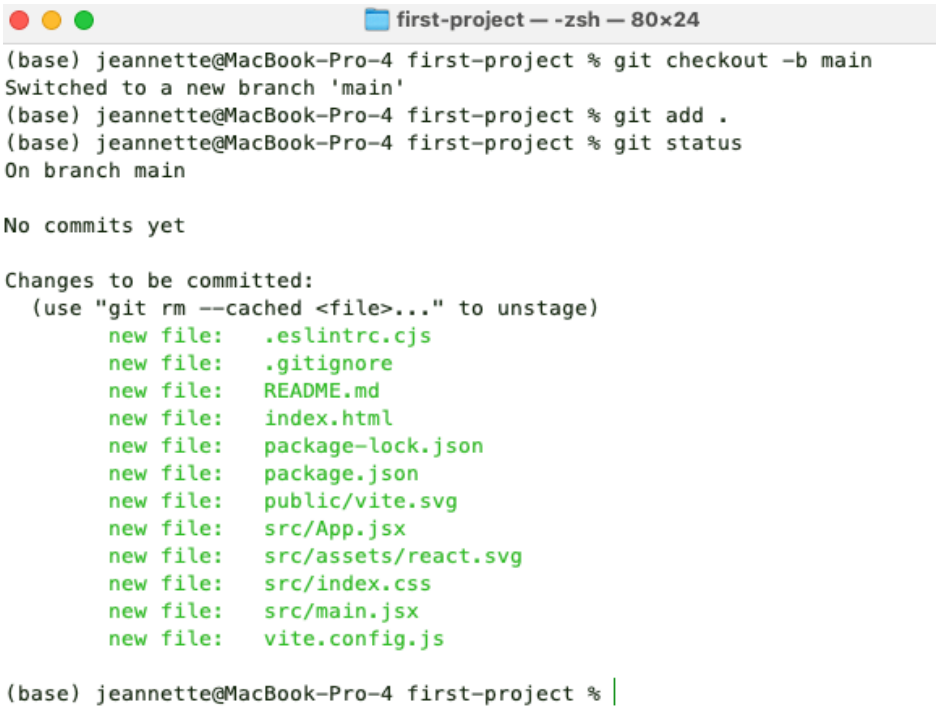
14. Next, add all the files by typing '`git add .`':



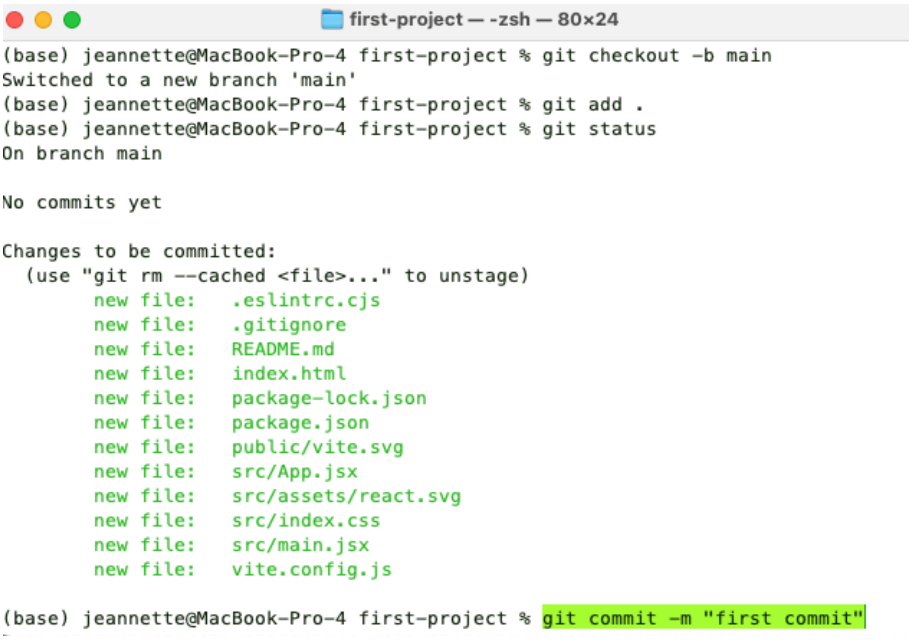
15. You can check the status by typing '`git status`' command:



16. As you will see from the output, `git` has added all the files, except 'node_modules' folder (because it is written in my `.gitignore` file), to the staging area:



17. To commit, we type the '`git commit`' command follow by a message, as follows:

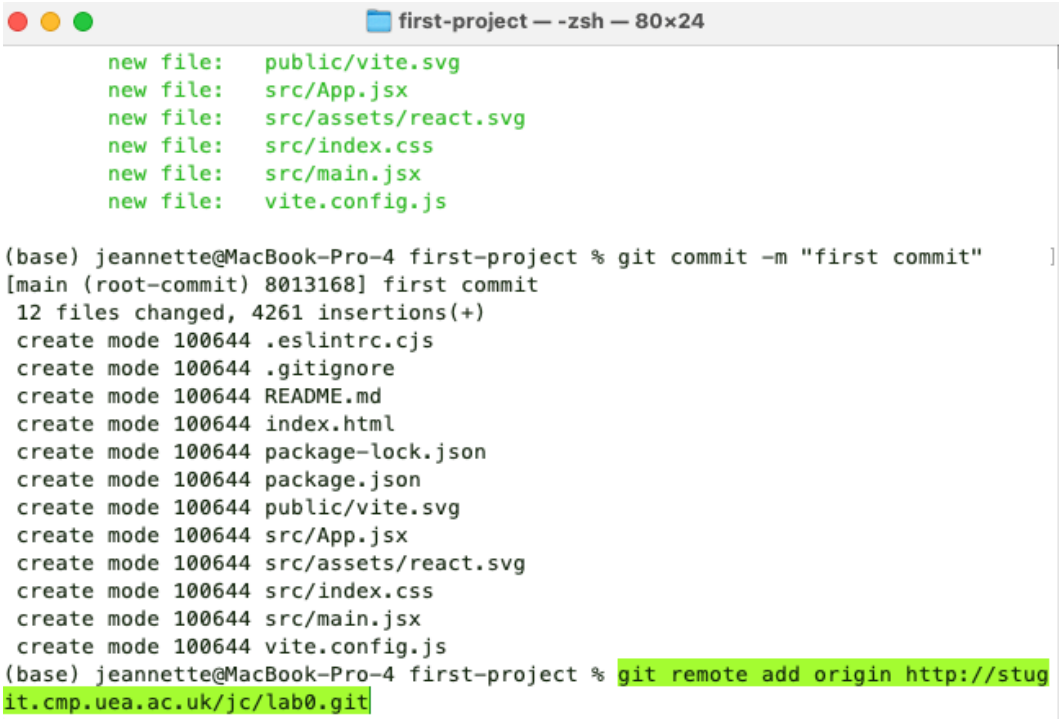


18. Here, you can check the files are committed to the local repo, for me, I have 12 files committed:

```
(base) jeannette@MacBook-Pro-4 first-project % git commit -m "first commit"
[main (root-commit) 8013168] first commit
 12 files changed, 4261 insertions(+)
 create mode 100644 .eslintrc.cjs
 create mode 100644 .gitignore
 create mode 100644 README.md
 create mode 100644 index.html
 create mode 100644 package-lock.json
 create mode 100644 package.json
 create mode 100644 public/vite.svg
 create mode 100644 src/App.jsx
 create mode 100644 src/assets/react.svg
 create mode 100644 src/index.css
 create mode 100644 src/main.jsx
 create mode 100644 vite.config.js
```

19. Now we would want to push the files to remote server – our CMP git server. First, add the remote server by typing:
`git remote add origin http://stugit.cmp.uea.ac.uk/jc/lab0.git`

Note: you will need to replace the URL with your own one (i.e. the one you copied in Task 7 above)



20. Now push the files by typing "`git push -u origin main`":

```
(base) jeannette@MacBook-Pro-4 first-project % git commit -m "first commit"
[main (root-commit) 8013168] first commit
 12 files changed, 4261 insertions(+)
 create mode 100644 .eslintrc.cjs
 create mode 100644 .gitignore
 create mode 100644 README.md
 create mode 100644 index.html
 create mode 100644 package-lock.json
 create mode 100644 package.json
 create mode 100644 public/vite.svg
 create mode 100644 src/App.jsx
 create mode 100644 src/assets/react.svg
 create mode 100644 src/index.css
 create mode 100644 src/main.jsx
 create mode 100644 vite.config.js
(base) jeannette@MacBook-Pro-4 first-project % git remote add origin http://stug
it.cmp.uea.ac.uk/jc/lab0.git
(base) jeannette@MacBook-Pro-4 first-project % git push -u origin main
```

21. Your will prompt for your username:

```
(base) jeannette@MacBook-Pro-4 first-project % git commit -m "first commit"
[main (root-commit) 8013168] first commit
 12 files changed, 4261 insertions(+)
 create mode 100644 .eslintrc.cjs
 create mode 100644 .gitignore
 create mode 100644 README.md
 create mode 100644 index.html
 create mode 100644 package-lock.json
 create mode 100644 package.json
 create mode 100644 public/vite.svg
 create mode 100644 src/App.jsx
 create mode 100644 src/assets/react.svg
 create mode 100644 src/index.css
 create mode 100644 src/main.jsx
 create mode 100644 vite.config.js
(base) jeannette@MacBook-Pro-4 first-project % git remote add origin http://stug
it.cmp.uea.ac.uk/jc/lab0.git
(base) jeannette@MacBook-Pro-4 first-project % git push -u origin main
Username for 'http://stugit.cmp.uea.ac.uk':
```

22. Enter the one that has been assigned to you. Mine is called 'jc'

```
(base) jeannette@MacBook-Pro-4 first-project % git commit -m "first commit"
[main (root-commit) 8013168] first commit
 12 files changed, 4261 insertions(+)
 create mode 100644 .eslintrc.cjs
 create mode 100644 .gitignore
 create mode 100644 README.md
 create mode 100644 index.html
 create mode 100644 package-lock.json
 create mode 100644 package.json
 create mode 100644 public/vite.svg
 create mode 100644 src/App.jsx
 create mode 100644 src/assets/react.svg
 create mode 100644 src/index.css
 create mode 100644 src/main.jsx
 create mode 100644 vite.config.js
(base) jeannette@MacBook-Pro-4 first-project % git remote add origin http://stug
it.cmp.uea.ac.uk/jc/lab0.git
(base) jeannette@MacBook-Pro-4 first-project % git push -u origin main
Username for 'http://stugit.cmp.uea.ac.uk': jc
```

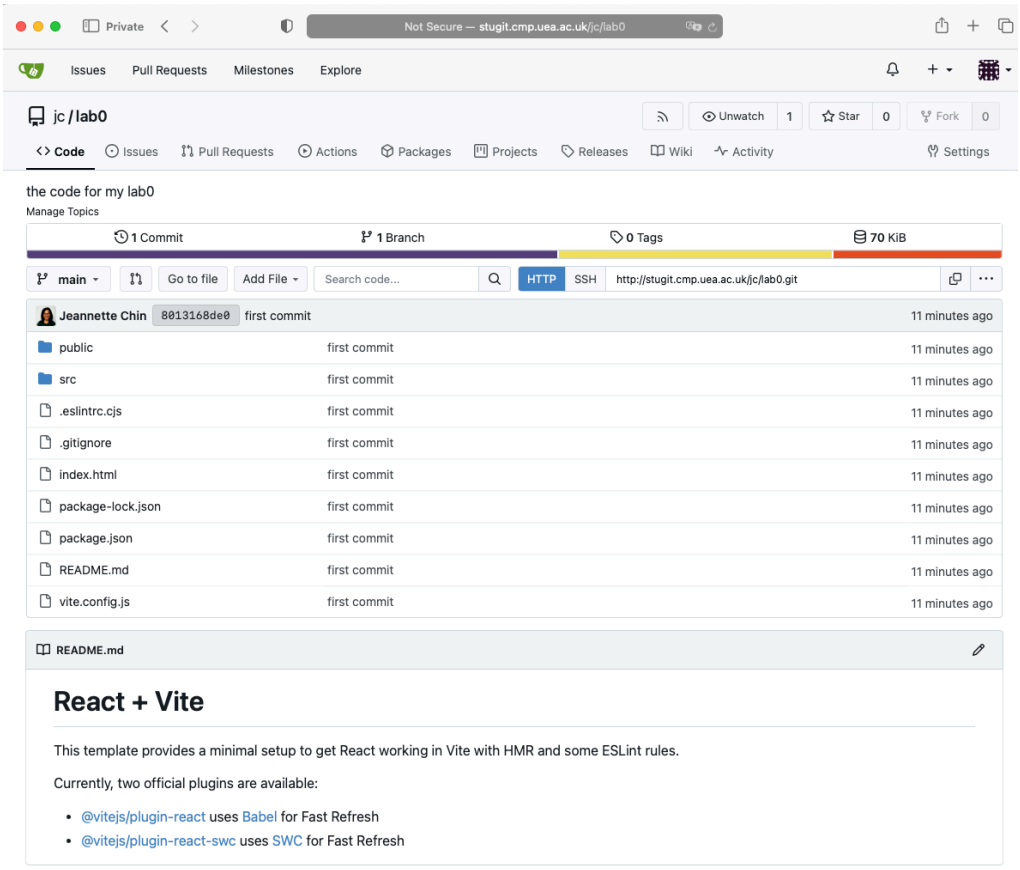
23. Enter the password as well when prompt. **Note you won't see your password on the terminal window**, just enter after you type/paste in your password.

```
(base) jeannette@MacBook-Pro-4 first-project % git commit -m "first commit"
[main (root-commit) 8013168] first commit
 12 files changed, 4261 insertions(+)
 create mode 100644 .eslintrc.cjs
 create mode 100644 .gitignore
 create mode 100644 README.md
 create mode 100644 index.html
 create mode 100644 package-lock.json
 create mode 100644 package.json
 create mode 100644 public/vite.svg
 create mode 100644 src/App.jsx
 create mode 100644 src/assets/react.svg
 create mode 100644 src/index.css
 create mode 100644 src/main.jsx
 create mode 100644 vite.config.js
(base) jeannette@MacBook-Pro-4 first-project % git remote add origin http://stug
it.cmp.uea.ac.uk/jc/lab0.git
(base) jeannette@MacBook-Pro-4 first-project % git push -u origin main
Username for 'http://stugit.cmp.uea.ac.uk': jc
Password for 'http://jc @stugit.cmp.uea.ac.uk':
```

24. Once successful, you should see the output similar to below:

```
(base) jeannette@MacBook-Pro-4 first-project % git push -u origin main
Username for 'http://stugit.cmp.uea.ac.uk': jc
Password for 'http://jc @stugit.cmp.uea.ac.uk':
Enumerating objects: 17, done.
Counting objects: 100% (17/17), done.
Delta compression using up to 10 threads
Compressing objects: 100% (14/14), done.
Writing objects: 100% (17/17), 39.49 KiB | 9.87 MiB/s, done.
Total 17 (delta 0), reused 0 (delta 0), pack-reused 0
remote: . Processing 1 references
remote: Processed 1 references in total
To http://stugit.cmp.uea.ac.uk/jc/lab0.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
(base) jeannette@MacBook-Pro-4 first-project %
```

25. To verify this, refresh CMP git web page and you should see the files are committed to this remote server:



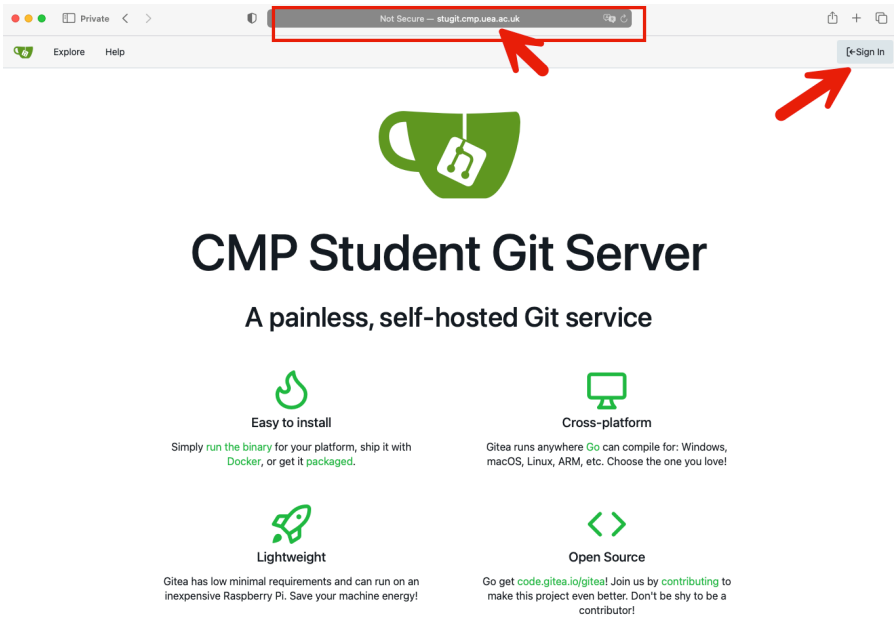
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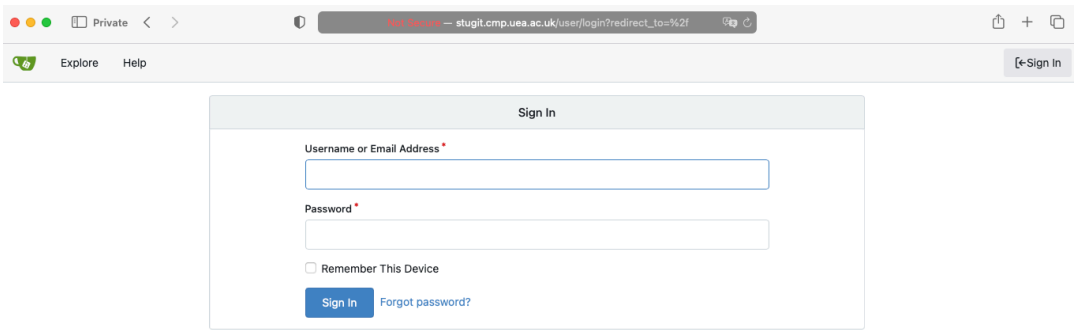
Setting up a Group Repo

This will be needed for both assignments. Below is the guidance on how to set up a group repo. Between your group, decide who is the lead. The lead will create this group repo.

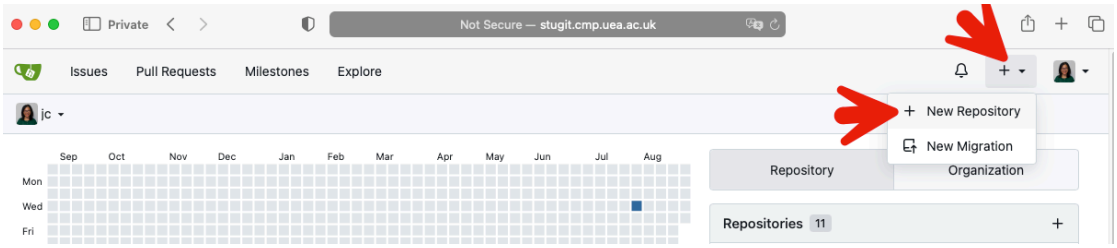
Steps:

1. On CMP git server, login

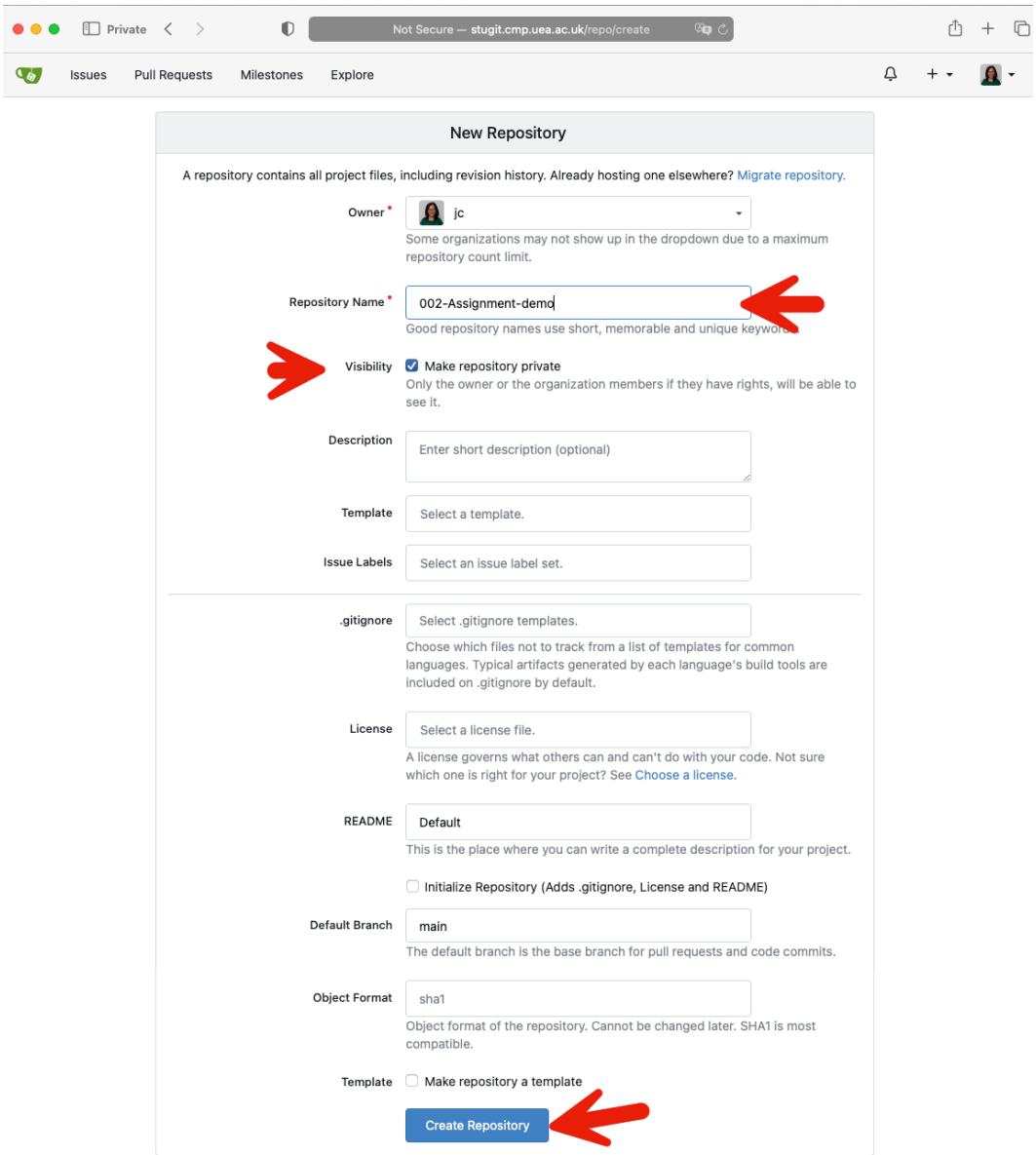




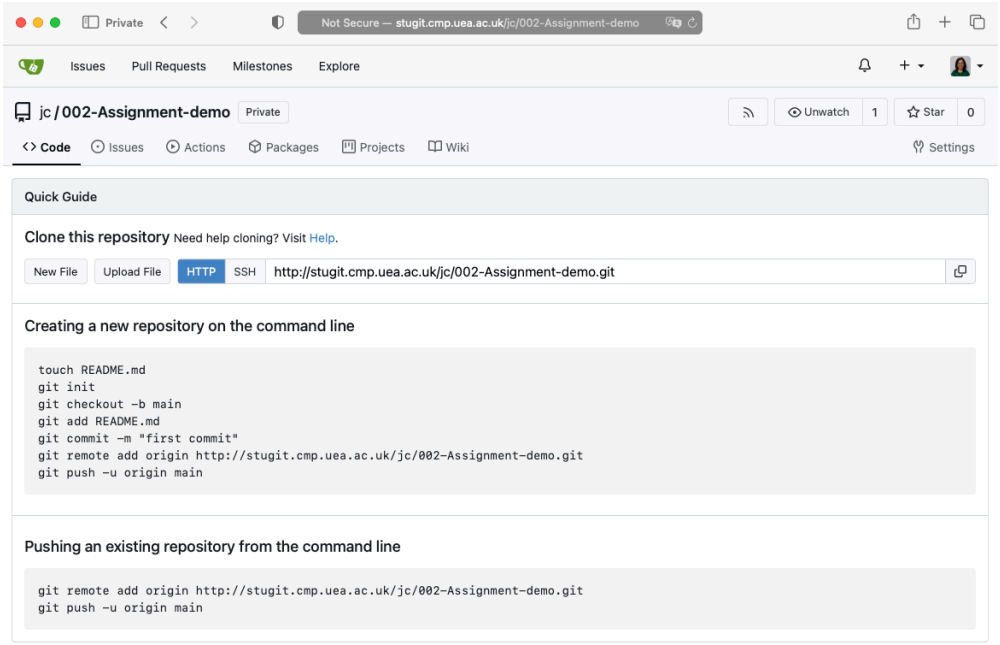
2. Create a repo for assignment (each assignment requires a repo)



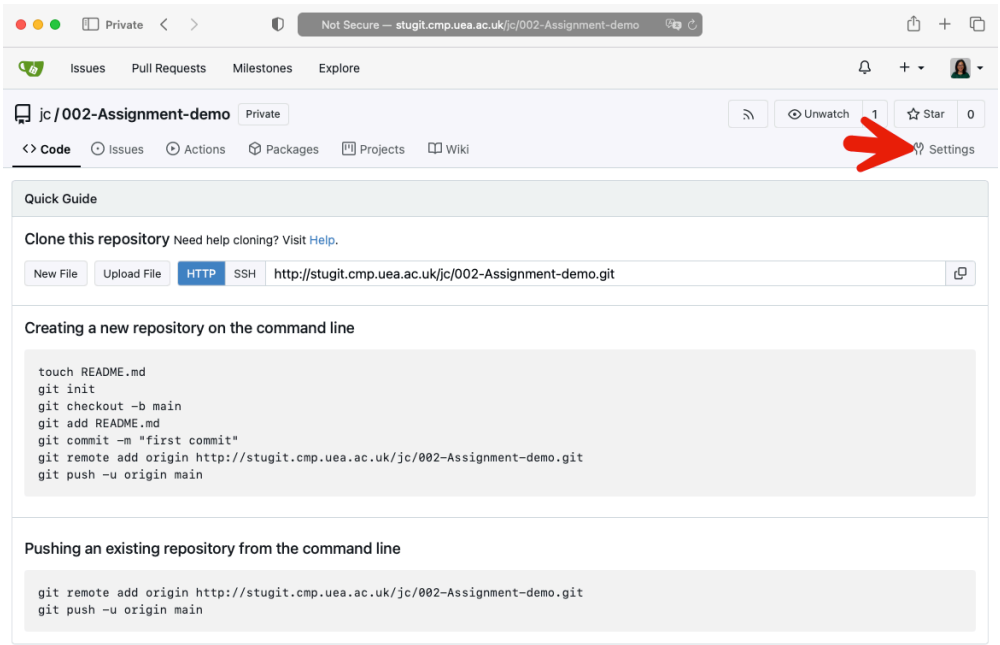
3. Give the new repo a name, by default the repo is private:



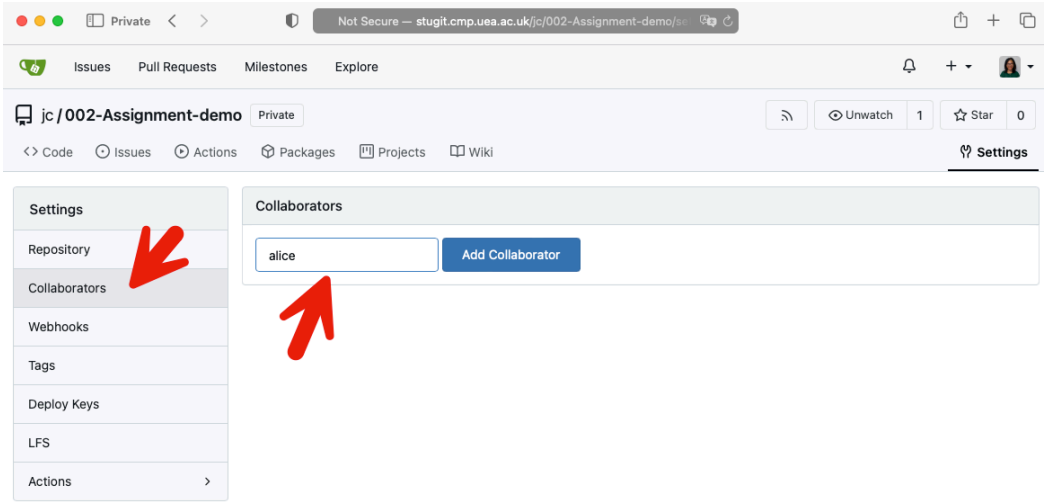
4. Now check the new repo on your main page, here is mine for your reference:



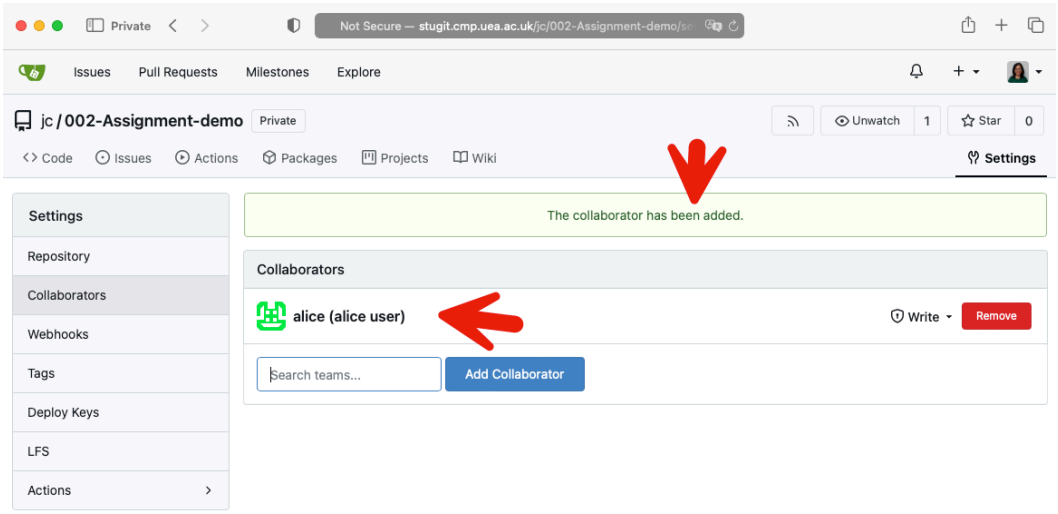
5. Go to 'Settings' :



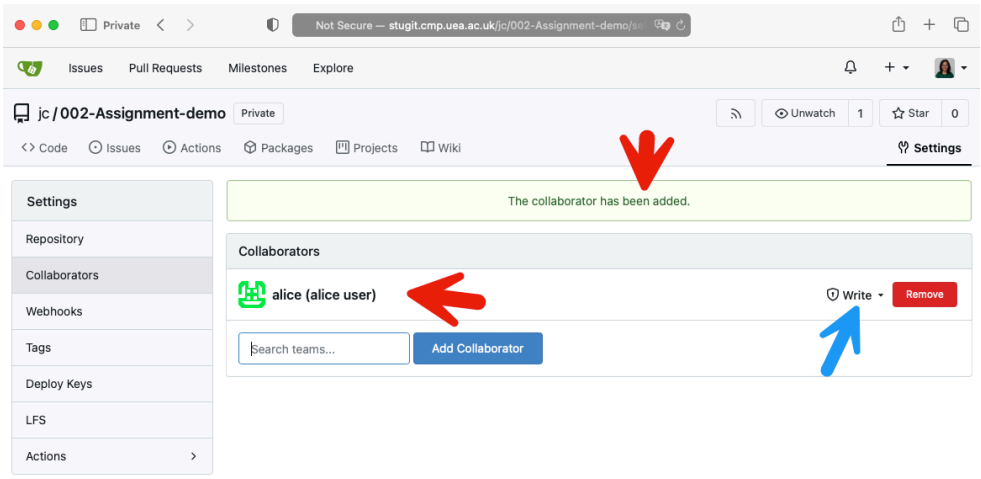
6. On the “Collaborators” section, type in your team member in the search box. You may need to ask your team members for their username.



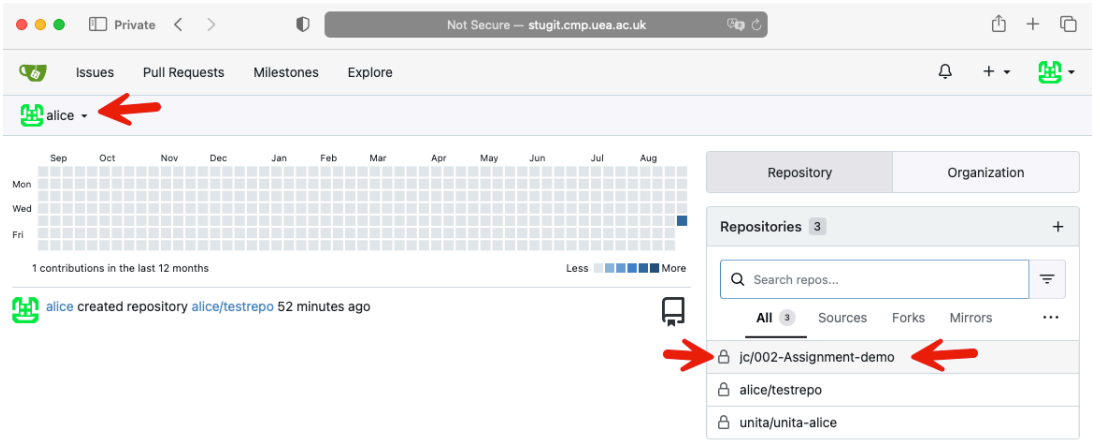
7. Once the system found the user, it will add them to your repo as a collaborator.



8. By default, your collaborators will be given ‘write’ permission.



9. Once you have done this, ask your team member to login to their account. They should be able to see your repo on their main page. Below is my user Alice’s account page, you can see my repo is displayed on her landing page, once I have added her in step 8 above:



10. Finally, add me to your group repo. My username is ‘jc’. (it is important to add me as a collaborator there). Dr Cheng Wang’s account information will be advised nearer the time.

When you reach to this point, well done!

For git command and usage go to : https://education.github.com/git-cheat-sheet-education.pdf